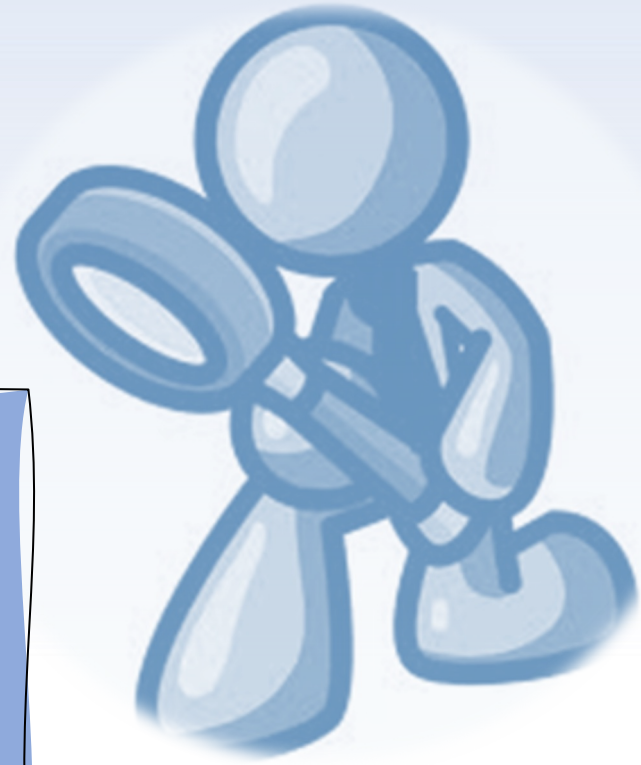




GOOD MORNING

Diagnostic Skills-
Intra Oral
Examination



Dr. Vanaja Reddy
Associate professor
Oral medicine & Radiology

Lecture Learning Outcomes

- **L9: Basic Diagnostic Skills: Intra oral Examination**
- **CLO LLO By the end of this lecture, students should be able to:**
- 2.6 L9.1 Enumerate the components of Intra oral examination in the dental case sheet
- 2.6 L9.2 Explain the importance of Intra oral examination for dental patients

COMPONENTS OF CASE HISTORY

The Diagnostic Process : *a systematic collection of data leading to identification of disease*

I. Taking & Recording of Patient's History

II. Examination

III. Provisional Diagnosis

IV. Investigations

V. Final Diagnosis



II. EXAMINATION

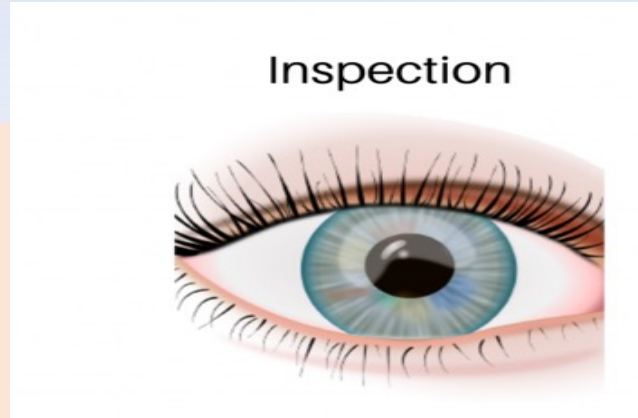
A. General

B. Extra-oral

C. Intra-oral :

Principles Of Examination

1. Inspection



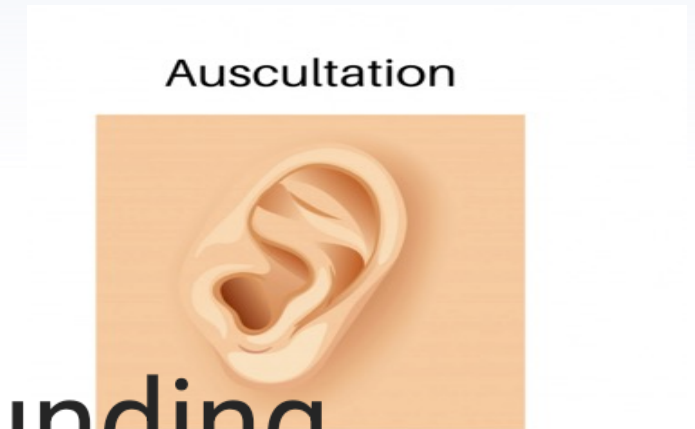
2. Palpation



3. Percussion



4. Auscultation



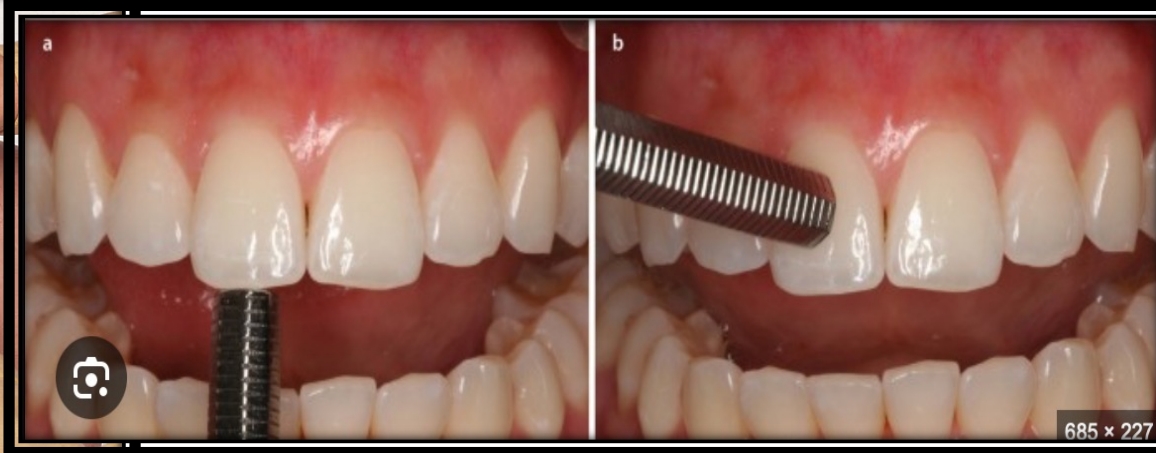
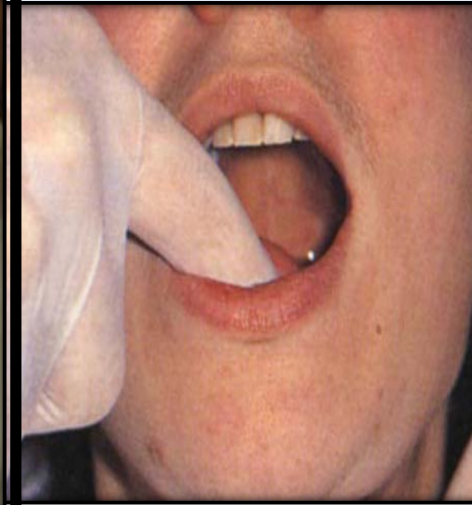
Number size shape and surrounding

Principles Of Examination

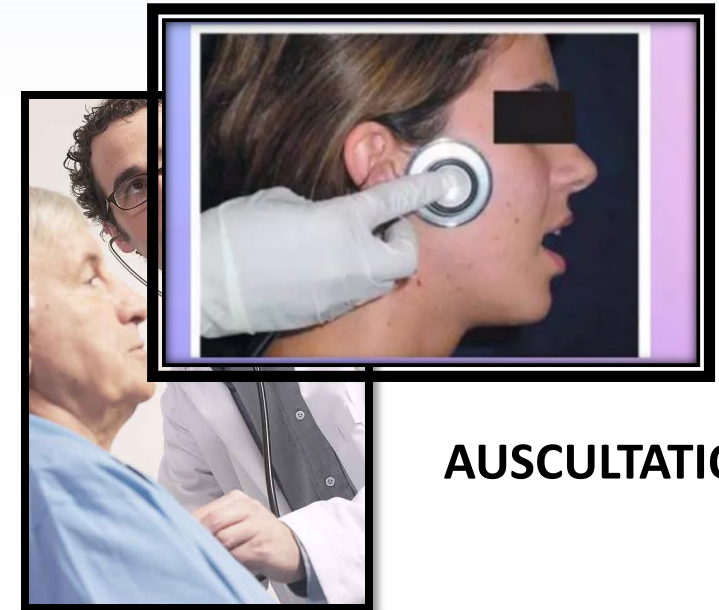
INSPECTION



PALPATION



PERCUSSION



AUSCULTATION

II. EXAMINATION

Extra-oral Examination

Intra-oral Examination

Soft tissue Examination

Hard tissue Examination

Gingiva
Periodontium
Mucosa

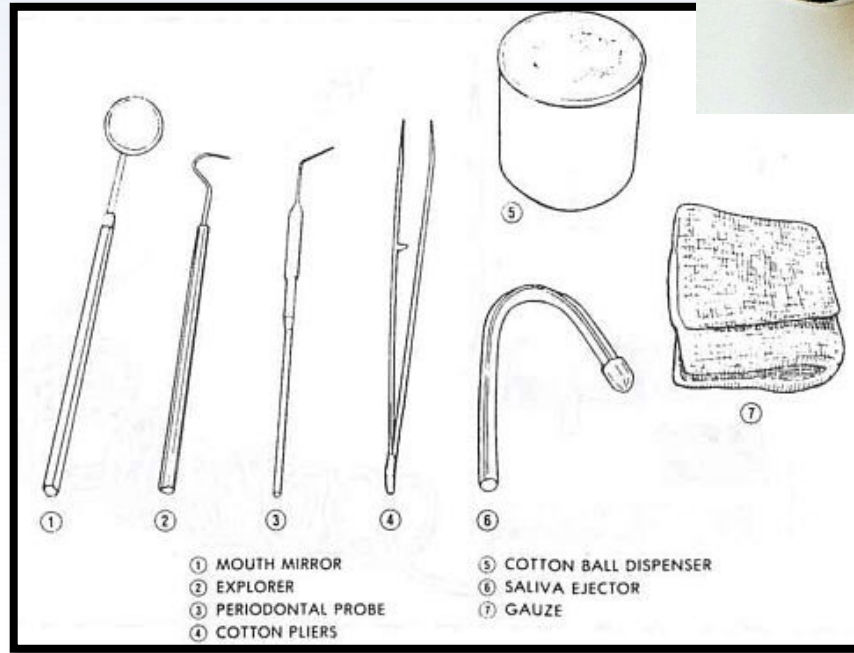
Teeth
Dental charting
Occlusion
Wasting disease

Requirements for Charting

■ A Diagnostic kit which contains:

1. Mouth mirror
2. Straight probe
3. Tweezer
4. Explorers
5. Periodontal probe
6. Gauze sponges
7. Saliva Ejector
8. Metal scale

■ Color Pencils/Pens: **BLUE**, **RED** and **BLACK**



II. EXAMINATION -

INTRAORAL EXAMINATION

SOFT TISSUE EXAMINATION-MUCOSA



Labial mucosa



Buccal mucosa



Parotid duct



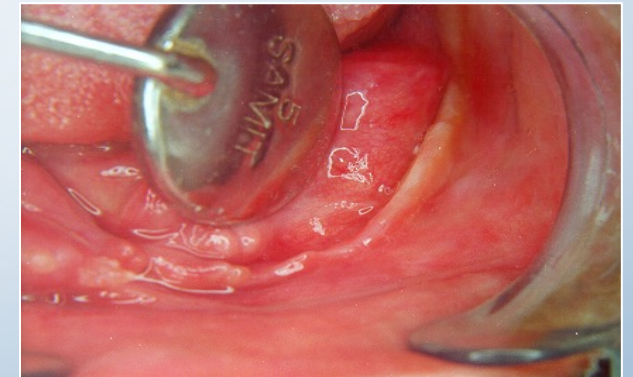
palate



Uvula, tonsils



tongue



vestibule



Frenal attachment

Examination of Labial and Buccal Mucosa

Examination of Lip:

Note lip color, texture, Vermilion borders, continues or not

Record any ulcer, fissure and any surface abnormalities

Observe:

Color [red, pink], Texture

Swelling

Ulcers

Minor salivary glands

Report any surface abnormalities



Examination of the Tongue

Shape, color, texture

Swellings

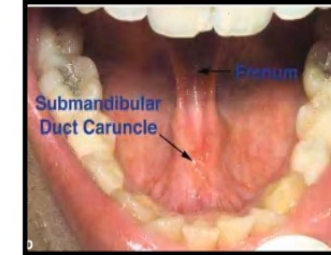
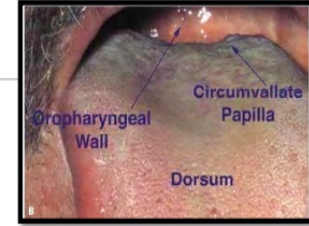
Ulcers

Tongue size

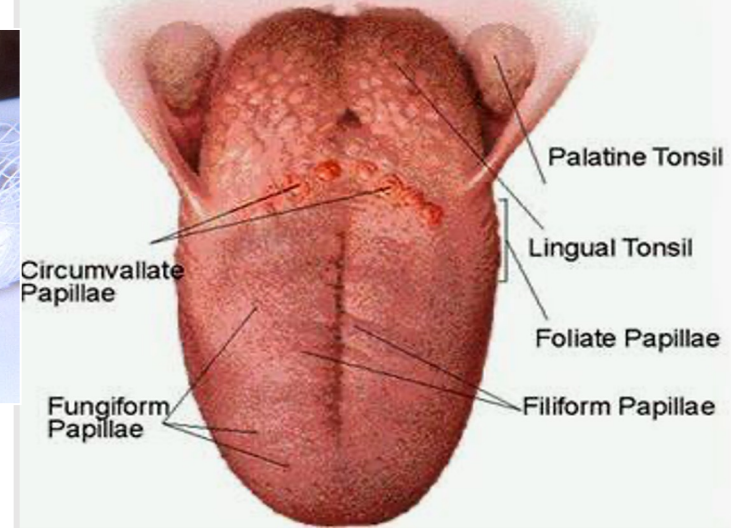
Taste buds

Functional deviation

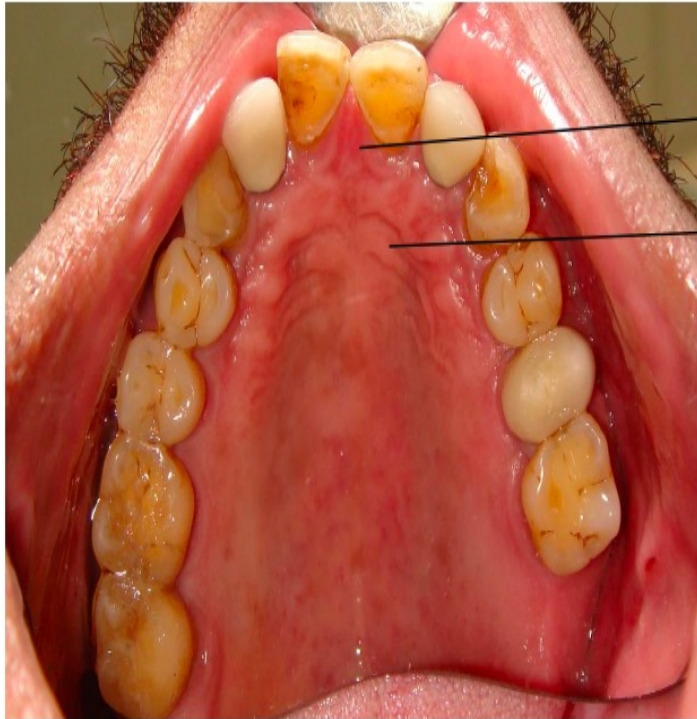
Scalloping



Tongue



Examination of Hard and Soft Palate



→ Incisive papilla

→ Palatal Rugae

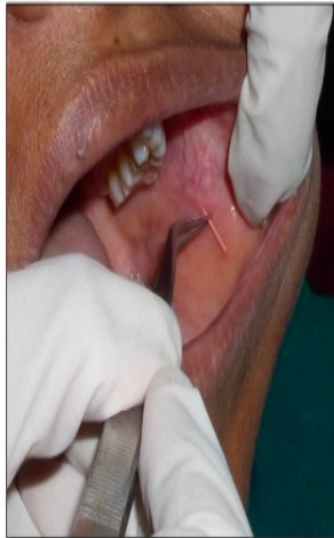
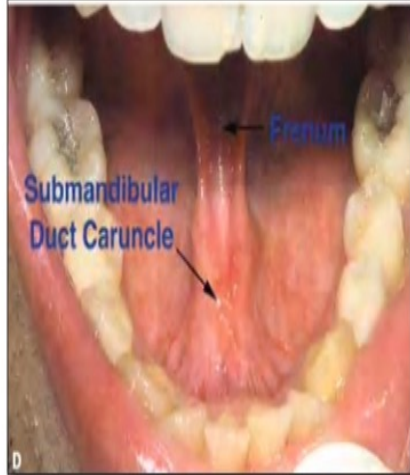
Soft Palate and Uvula

Soft palate: check for color, size shape, petechiae, ulcer, trauma, minor salivary gland openings, gag reflex

Uvula: size, shape, midline location



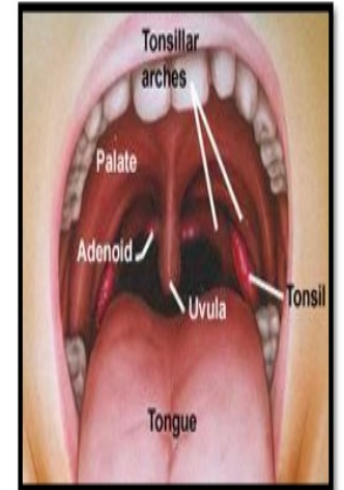
Salivary Gland Duct Orifice



Examination of Oropharynx

Techniques of Examination:

- Depress the tongue with Mouth mirror/tongue depressor(wooden spatula)
- Retracting the tongue manually using gauze



Tonsils: Examine color, enlargement, surface character

II. EXAMINATION -

INTRAORAL EXAMINATION

SOFT TISSUE EXAMINATION. - GINGIVA AND PERIODONTAL STATUS

SIZE



COLOUR



CONTOUR



CONSISTENCY



TEXTURE

BLEEDING ON PROBING



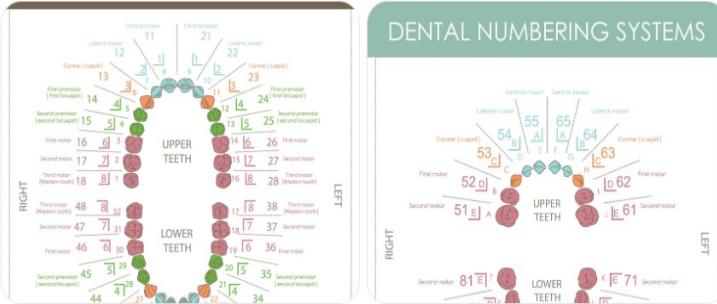
POCKETS



II. EXAMINATION -

INTRAORAL EXAMINATION -TEETH

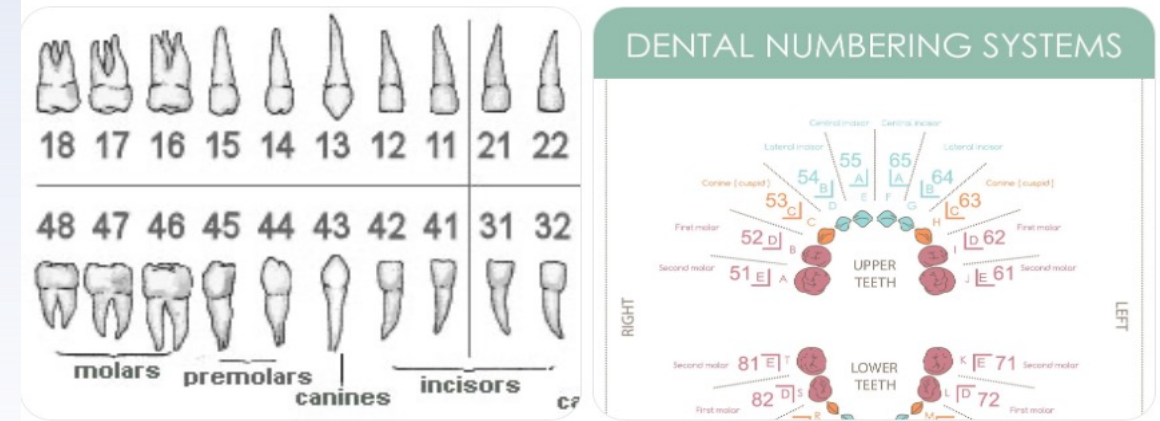
1. Universal Numbering System (ADA System)



Permanent Dentition

- Teeth are numbered 1–32
- Starts from **maxillary right third molar (1)** → across → **maxillary left third molar (16)**
- Continues **mandibular left third molar (17)** → across → **mandibular right third molar (32)**

2. FDI Two-Digit System (ISO System)



Structure

- First digit** → Quadrant
- Second digit** → Tooth position from midline

TEETH NOMENCLATURE-

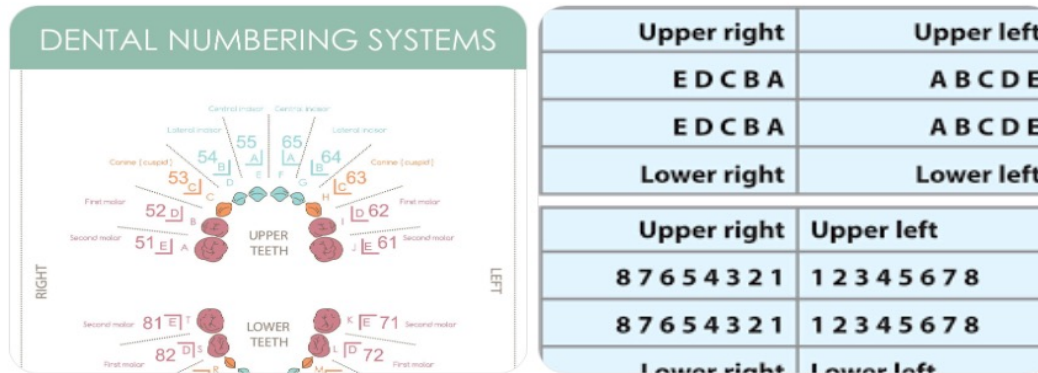
F.D.I system. 18 17 16 15 14 13 12 11 21 22 23 24 25 26 27 28
48 47 46 45 44 43 42 41 31 32 33 34 35 36 37 38

DECIDUOUS TEETH - 55 54 53 52 51 61 62 63 64 65
85 84 83 82 81 71 72 73 74 75

II. EXAMINATION -

INTRAORAL EXAMINATION -TEETH

3. Palmer Notation System (Zsigmondy–Palmer)



Permanent Dentition

- Teeth numbered **1–8** from midline
- Quadrants indicated by **L-shaped symbols**

II. EXAMINATION -

INTRAORAL EXAMINATION -TEETH

TEETH NOMENCLATURE-

F.D.I system. 18 17 16 15 14 13 12 11 21 22 23 24 25 26 27 28
48 47 46 45 44 43 42 41 31 32 33 34 35 36 37 38

DECIDUOUS TEETH - 55 54 53 52 51 61 62 63 64 65
85 84 83 82 81 71 72 73 74 75

- # Type of dentition
- # Teeth present & teeth absent
- # Occlusion - Angle's classification & tooth position
- # Developmental disorders - no, shape, size, structure
- # Dental caries
- # Attrition, abrasion, erosion & fracture
- # Stains & discoloration
- # Mobility

Dental and Periodontal Charting

- Is a diagrammatic representation of dental findings. It is a quick, efficient, and accurate method of recording the large volume of detailed information that is needed to reflect the patient's dental status

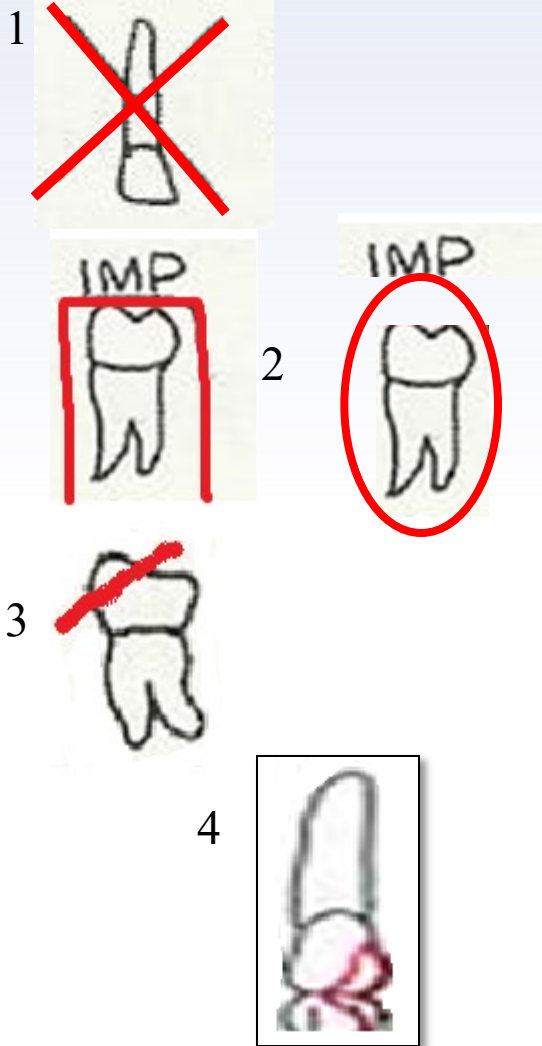
- It is a combination of COLOR CODED SYMBOLS

CLINICAL DENTAL FINDINGS																												PATIENT NUMBER									
4 3 4 3 3 4														3 2 2 2 2 2 1 2 1 2														3 2 3 3 2 3 3 3 4					3				
18 17 16 15 14 13 12 11														21 22 23 24 25 26 27 28														DATE									
																												FACIAL									
																												LINGUAL									
																												DATE									
4 3 5 5 3 4														3 2 3 2 2 3 2 2 2 2 2 2 2 2 2 2 3														3 3 2 3 2 3 3 3 4					1				
32 31 30 29 28 27 26 25														24 23 22 21 20 19 18 17														DATE									
																												FACIAL									
																												LINGUAL									
																												DATE									
3 2 3 2 1 2														2 1 3 2 1 2 2 1 2 2 2 2 2 2 2 2 3														3 2 3 2 3 2 3 2 3					1				
48 47 46 45 44 43 42 41														31 32 33 34 35 36 37 38														DATE									
																												FACIAL									
																												LINGUAL									
																												DATE									

Color Code in Dental and Periodontal Charting

- Blue, Red and Black colors are used in charting
- BLUE: Conditions NOT requiring treatment or intervention
- RED: Indicates intervention [further investigation/treatment] required
- BLACK color: Used for Gingival line

Charting Symbols



Common Findings:

1. Missing tooth (needs intervention- red .
- If blue –needs no intervention)

2. Impacted tooth

2. Partial eruption

3. Fractured tooth

Missing Teeth

- Third molar

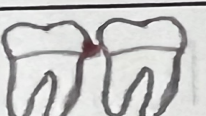
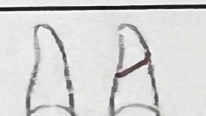
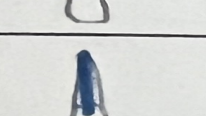
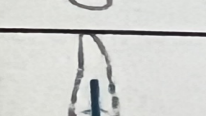
I. Completely invisible:

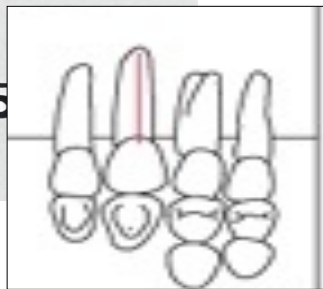
- If history of extraction (any age): X in BLUE
- If no history of extraction
 - Below the age of 21yrs: X mark with RED
 - Above the age of 21yrs: X mark with RED

II. Partially visible tooth:

- Below the age 21yrs: partially erupted tooth with RED
- Above the age of 21yrs: Impacted tooth with RED

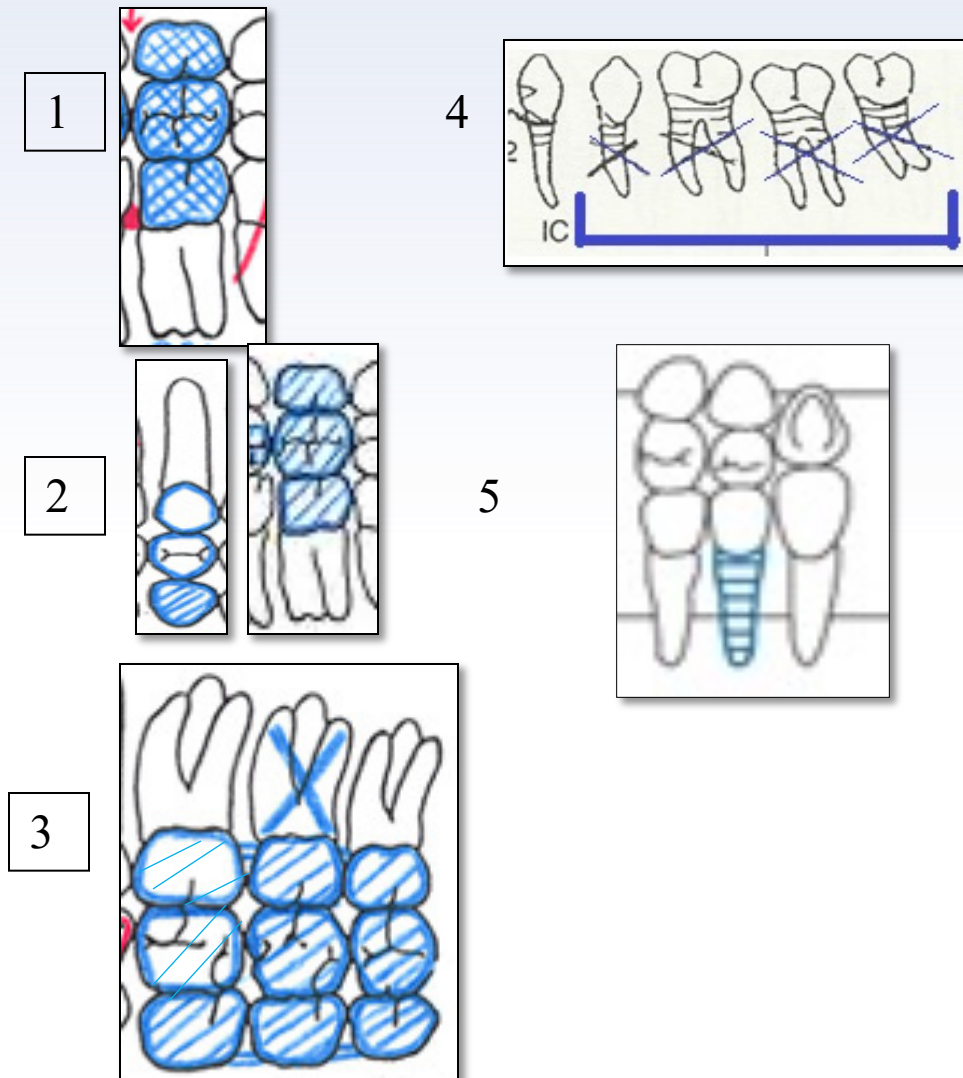
III. Premolar extracted for orthodontic treatment: X in BLUE

1		DENTAL CARIES (Red)
2		AMALGAM RESTORATION (Blue)
3		COMPOSITE RESTORATION (Blue outline only)
4		RESTORATION THAT NEEDS INTERVENTION (blue with red Outlin)
5		OVERHANGING RESTORATION (Red)
6		CROWN FRACTURE (Red) ROOT FRACTURE (Red)
7		PERIAPICAL PATHOLOGY (Red) [radiographic]
8		ROOT CANAL FILLED TOOTH (Blue)
9		INCOMPLETE RO CANAL FILLING (Blue)



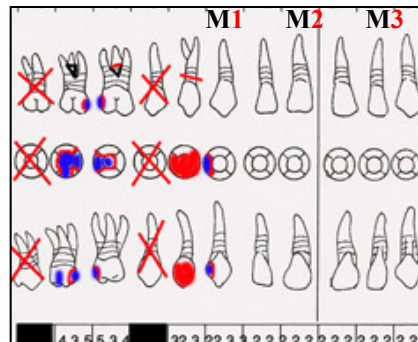
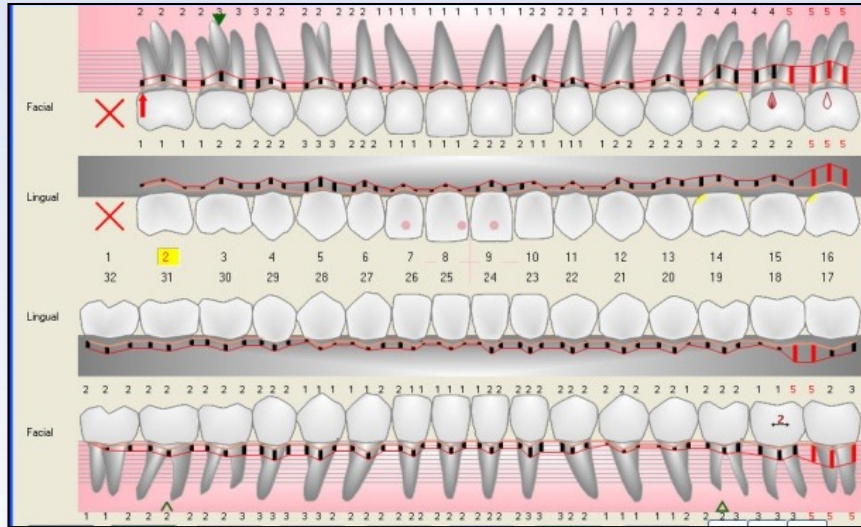
Restorative and Endodontic Findings:

1. Dental caries- **RED**
2. Amalgam Restoration: outline, then shade in **blue**
3. Tooth-colored restoration: **outline in blue**, do not shade[e.g. Composite restoration]
4. Restorations present but need to be replaced: color in solid blue and **outlines in red**
5. Over hanging restorations- **Red**
6. Crown fracture , root fracture -**Red**
7. Periapical pathology –**Red** (only after seeing radiograph)
8. Root canal filled- **BLUE**
9. Incomplete root canal filled-**BLUE**
- 10 Tooth that needs root canal therapy(**Red line**)



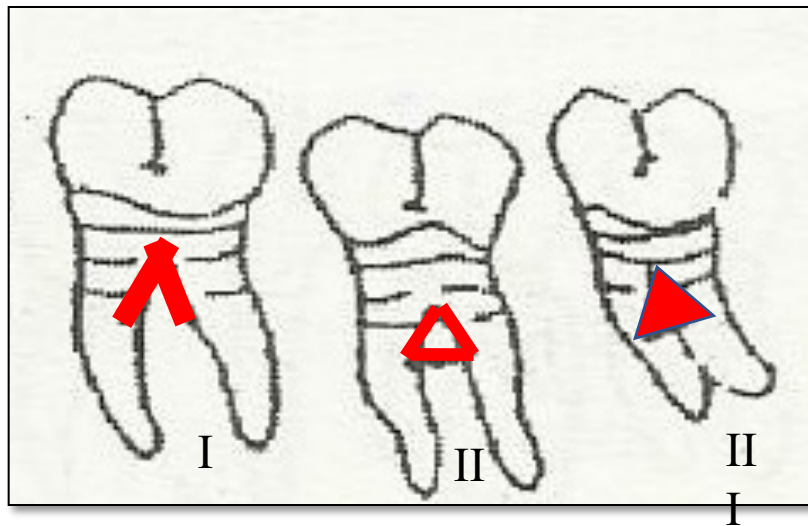
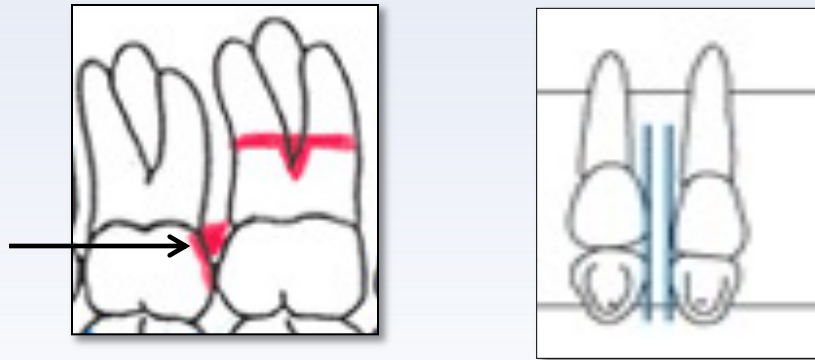
Prosthetic Findings:

1. Stainless crown: **Outline and hatchet - BLUE**
2. Porcelain fused to metal: outline and make diagonal stippling
3. **Fixed Bridge:** The root(s) in the pontic area(s) should be 'X'ed out and Connect abutments to pontic by horizontal lines
4. **Removable partial Denture:** The teeth in the pontic area(s) should be 'X'ed out. Draw a line at the apices connecting these areas and write, **all in blue**
5. Dental Implants -**blue**



Periodontal Findings in brief:

- Gingival margin [continuous black line], Mucogingival junction [Interrupted black line]
- Abnormal Gingival Margin Line and Recession: Draw a line in red indicating the position of the gingival margin, coronally if it is enlarged, or apically, if there is recession. Indicate in millimetres, the distance between the CEJ
- Tooth Mobility: Write symbol M₁ at the apex of the root and degree of mobility in red
 - M₁: Horizontal mobility 0.5-1 mm
 - M₂: Horizontal mobility, 1-2mm bucco-lingually
 - M₃: mobility in vertical direction or more than 2 mm horizontal mobility

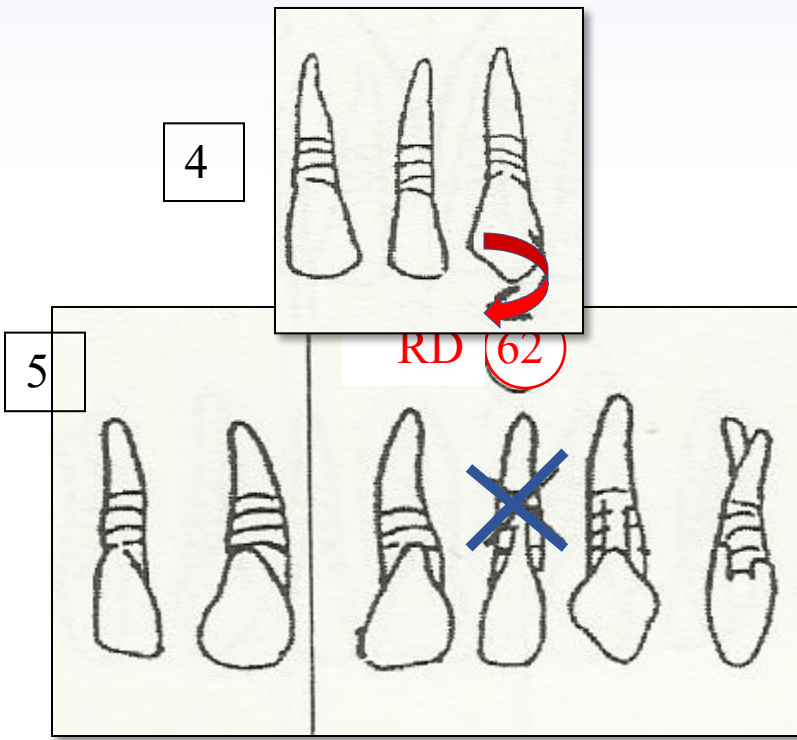
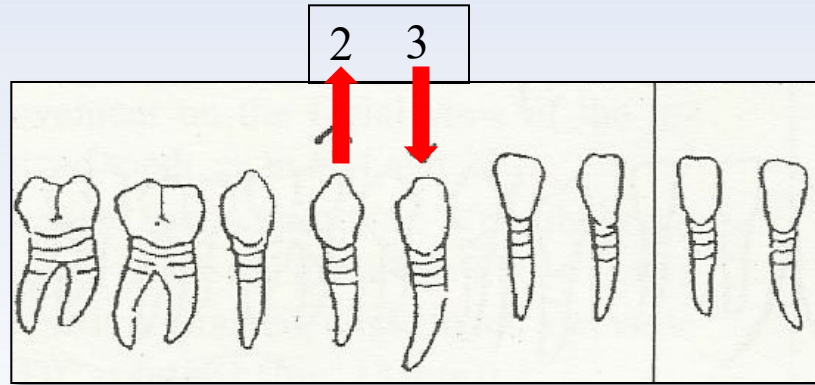


• Overhanging restoration- **RED**

• *Diastema* - **Blue**

• *Furcation involvement*: Indicate in **red** at the root and the sides of the teeth involved.

- I. Open V furcations is/are probable
- II. Open triangle furcation is probable under the furcation bridge
- III. Closed triangle through and through furcation involvement



• Findings related to eruption of teeth:

1. Partially erupted teeth
2. Over-erupted [Extrusion]: draw a arrow in **RED** in coronal direction ↑
3. Under-erupted teeth [Intrusion]: draw a arrow in **RED** pointing in apical direction ↓
4. Drifted, Rotated, Tipped teeth: draw curved arrow in **RED** ↻ showing the direction of rotation on facial view of the tooth
5. Retained Deciduous tooth: write tooth number of deciduous tooth in the region where it is retained, circle in **RED** and write [RD]

2.Percussion Test

- lateral/horizontal Percussion



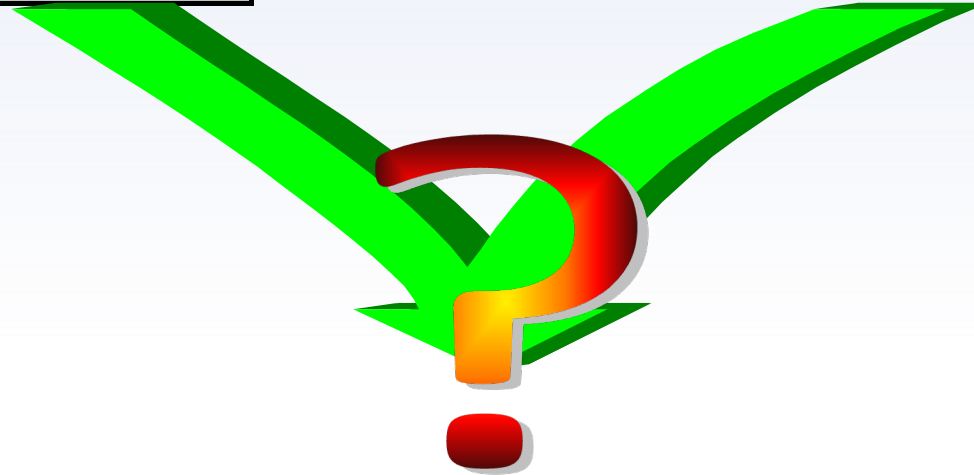
Vertical percussion

PATIENT	NUMBER
---------	--------

	MISSING TOOTH
	PERIAPICAL PATHOSIS
	ROOT CANAL FILLED
	INCOMPLETE ROOT CANAL FILLING
	SINUS TRACT
	IMPACTED
	PARTIAL ERUPTION
	INTRUSION
	EXTRUSION
	FRACTURE
	CARIES (Red)
	RESTORATION (Blue)
	GOLD CROWN
	GOLD CROWN
	ACRYLIC OR PORCELAIN FACING
	OVERHANG RESTORATION
	FOOD IMPACTION
	DRIFTING
	POOR CONTACT
	ROTATION
	FURCATION
	I
	II
	III

**I. Taking & Recording
of Patient's History**

II. Examination



III.

**Provisional/working/
clinical/tentative
Diagnosis**

- **Provisional diagnosis:** Is a process by which the practitioner distinguishes one diagnosis from another, differentiates between normal and abnormal.
- **Differential diagnosis :**Is a list of several disorders with similar histories and physical findings, from which the correct diagnosis is ultimately selected
- **Investigations**

A confirmed diagnosis based on all available data.-
history , examination, results of investigations.



FINAL DIAGNOSIS

Phases of Treatment Plan

- Emergency/preliminary phase
- **Phase I – Etiotropic (Initial / Cause-related) Therapy**

- Eliminate etiologic factors and control inflammation

- Patient education & motivation
- Plaque control instructions (brushing, interdental aids)
- Scaling and root planing (SRP)
- Removal/correction of local factors
(overhangs, faulty restorations, caries)
- Management of acute periodontal conditions
(periodontal abscess, ANUG)
- Control of systemic and environmental risk factors
(smoking cessation, diabetes control)
- Occlusal adjustment (if indicated)
- Provisional splinting of mobile teeth (if needed)

Newman MG, Takei HH, Klokkevold PR, Carranza FA.

Carranza's Clinical Periodontology, 14th Edit, 2023,. Elsevier

Phases of Treatment Plan

Phase II – Surgical Therapy

-Eliminate residual pockets and correct anatomical defects

Indications:

- Persistent pockets after Phase I
- Furcation involvement
- Osseous defects
- Gingival deformities

- {Periodontal flap surgery
- Osseous resective surgery
- Regenerative procedures
(bone grafts, GTR, biologics)
- Mucogingival surgery
(root coverage, gingival augmentation)
- Crown lengthening (when indicated)}

Phases of Treatment Plan

Phase III – Restorative / Corrective Therapy

-Restore function, esthetics, and stability

- {Permanent restorations
- Prosthodontic rehabilitation
- Endodontic therapy (if required)
- Orthodontic treatment (selected cases)
- Occlusal rehabilitation
- Replacement of missing teeth (bridges, implants)}

Phases of Treatment Plan

Phase IV – Supportive Periodontal Therapy (Maintenance Phase)

Prevent recurrence of disease

- {Periodic **recall visits** (3–6 months)}
- Reinforcement of oral hygiene
- Professional plaque and calculus removal
- Monitoring probing depth, attachment levels, mobility
- Management of recurrent disease}

Conclusion

- Thus successful treatment depends on :
- S: Subjective symptoms
- O: Objective symptoms
- A: Assessment when patient first comes to clinic
- P: Record of the procedures carried out

Reference and further reading:

1. Bricker and miller oral diagnosis and treatment planning 2nd edition pg 41-60
2. Text book of oral medicine -2nd edition –ghoms. Chap 7 pg-60.
3. Burket's Oral Medicine, 13th Edit,chap 1, pg 1-15.

Thank u